Do Latent Volatility Regimes Improve Risk-Based Asset Allocation?

Evidence from a Preregistered Walk-Forward ETF Study

Shakhrukh Kakhramonov*

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Abstract

Between 2010 and 2026 I ran a minimum-variance portfolio of five US exchange-traded funds that conditioned its covariance forecast on volatility states inferred in real time from a two-state Gaussian hidden Markov model, and compared it against the identical portfolio without regime conditioning. Net of 10 basis points in transaction costs, the regime-aware strategy earned an annualized Sharpe ratio 0.021 higher; the 95% stationary-bootstrap interval runs from −0.075 to 0.115, and the annualized return difference is 0.314 bps. Regime conditioning lowered annualized volatility by 0.175 pp, but required 3.2 times the turnover, and the estimate falls from 0.031 to 0.011 as costs rise from zero to 20 basis points. Across thirteen preregistered robustness specifications the sign reverses twice and no comparison survives multiplicity adjustment; portfolio constraints affected apparent performance more than any modelling choice examined. The hypothesis, sample split and robustness grid were registered before estimation.

Keywords: regime switching, hidden Markov models, minimum-variance portfolios, out-of-sample evaluation, preregistration, transaction costs

JEL: G11, G17, C58

1 Introduction

A minimum-variance portfolio of five US exchange-traded funds that conditioned its covariance forecast on volatility states inferred in real time earned an annualized Sharpe ratio 0.021 higher than the same portfolio without regime conditioning over 2010–2026, net of 10 basis points in transaction costs. The 95% confidence interval on that difference runs from −0.075 to 0.115.

*Baruch College, Zicklin School of Business. Code, preregistration and complete output: `https://github.com/kakhramonovsh-prog/regime-aware-asset-allocation`. I thank the reviewers who read successive drafts of the design and forced several corrections recorded in the amendment log.

The strategy did what the theory predicts, realizing 0.175 pp lower annualized volatility, but it traded 3.2 times as much as its comparator, and the estimated advantage falls monotonically as costs rise. On this evidence, regime conditioning delivers a measurable risk reduction too small relative to sampling variation to establish a risk-adjusted improvement once trading costs are paid.

The design was fixed before the data were touched. I published the hypothesis, the comparator, the sample split, the cost convention and all thirteen robustness specifications, and version-tagged them in a public repository before estimating a single model. Four later amendments each carry a date and a record of what was known when I made them; three were adopted before any portfolio return existed. The comparison reported here is the one committed to in advance, not the one that survived a search. Preregistration, frozen data, causal tests, automated validation and dated amendments substantially constrained ex-post researcher discretion, though they did not eliminate it.

The choice of comparator does most of the analytical work. Regime-aware minimum variance beats equal weight and 60/40 comfortably on realized Sharpe, and a study could stop there and report an improvement. It should not. Those benchmarks differ from the regime strategy in optimization, covariance estimation and regime conditioning at the same time, so a gap between them attributes nothing to any single ingredient. I therefore compare against rolling Ledoit–Wolf minimum variance, which shares every ingredient except the regime signal. The 0.021 difference measures what regime conditioning adds on top of dynamic covariance estimation, and it is not distinguishable from zero.

Three findings temper the point estimate further. The sign reverses under a three-state specification and when realized volatility is dropped from the feature set. It reverses again between the pre- and post-2020 halves of the sample. And after Holm adjustment across the robustness family, no specification retains significance. Set against these, a positive point estimate with an interval 9.0 times its width is best read as inconclusive about small effects rather than as evidence of either a benefit or its absence.

This paper contributes in three ways. It tests a regime-conditioning strategy under a preregistered design with a matched comparator, which is uncommon in the backtesting literature and directly addresses the multiple-testing critique. It reports the full ablation ladder, showing where risk-adjusted gains actually arise: mostly from optimization itself rather than from dynamic covariance or regime conditioning. And it documents an interaction the design did not anticipate, in which portfolio constraints moved the estimated difference more than any modelling choice examined.

2 Related literature

Three strands bear directly on this study.

Hamilton [1989] introduced the regime-switching framework this paper applies, and with it the discipline that states are latent and inferred rather than observed. That distinction governs the empirical design here: the trading signal is the filtered probability at the decision date, never a smoothed path that conditions on data the investor did not have. The volatility clustering the framework describes was documented by Engle [1982] and generalized by Bollerslev [1986]; this paper does not revisit it, asking instead whether acting on it survives out-of-sample testing net of costs.

Applying regime switching to allocation has a substantial literature. Ang and Bekaert [2002] document regime shifts in international equity correlations and their allocation consequences, Guidolin and Timmermann [2007] solve multivariate regime-switching allocation problems, and Ang and Timmermann [2012] survey the field. This paper's contribution is narrower and more sceptical than a survey of that work would suggest: it holds three things fixed at once — states inferred strictly in real time, a comparator differing by the regime signal alone, and trading costs charged on realized turnover — and preregisters the comparison before estimating anything. The question is not whether regimes exist or whether conditioning on them helps in principle, but what the conditioning is worth after those three constraints are imposed simultaneously.

DeMiguel et al. [2009] provide the closest methodological precedent. They evaluate fourteen portfolio models across seven datasets and find that none consistently outperforms naive 1/N allocation on Sharpe ratio, certainty-equivalent return or turnover, concluding that estimation error more than offsets the theoretical gain from optimization. Their framing is the model for this study: a negative out-of-sample result becomes informative when it identifies the mechanism by which a theoretically superior method fails. The question here is one rung further along — optimization is granted, and the test is whether regime conditioning adds anything on top of it.

Harvey and Liu [2015] and Harvey et al. [2016] supply the inferential standard. Documenting that hundreds of factors have been tested in the cross-sectional literature, they argue that conventional significance thresholds no longer establish anything and propose haircuts for reported Sharpe ratios that account for the number of trials. Preregistering a single primary comparison and applying Holm adjustment across an explicitly bounded robustness family is a direct response to that critique: the number of trials is fixed and published in advance rather than reconstructed afterward.

The portfolio problem is Markowitz [1952] with the mean removed. Michaud [1989] sets out why sample mean-variance optimization performs poorly out of sample, which is the reason no strategy here uses expected-return forecasts, and Clarke et al. [2011] characterize what minimum-variance portfolios actually hold. Jagannathan and Ma [2003] show that imposing constraints which are wrong as beliefs can nonetheless reduce estimation error and improve realized risk — a result that bears directly on the constraint interaction reported in Section 8.

On methods, the covariance estimator follows Ledoit and Wolf [2004], whose shrinkage approach is designed for exactly the near-collinear setting observed here, where the equity block carries pairwise correlations between 0.82 and 0.92. The hidden Markov model is estimated by the EM algorithm of Dempster et al. [1977]. Volatility forecast evaluation uses QLIKE following Patton [2011], who shows that rankings can be distorted when noisy volatility proxies are evaluated under inappropriate loss functions, with pairwise comparisons following Diebold and Mariano [1995]. Inference uses the stationary bootstrap of Politis and Romano [1994], chosen because portfolio returns are serially dependent and independent resampling would understate uncertainty; Ledoit and Wolf [2008] develop the robust Sharpe-ratio test that motivates bootstrapping the difference rather than comparing ratios directly. Standard errors on mean differences use Newey and West [1987], and multiplicity is handled by Holm [1979]. Bailey et al. [2014] give the sharpest statement of why backtested performance requires this kind of discipline.

3 Data and information availability

The universe is five liquid US-listed exchange-traded funds spanning equities, duration and gold: SPY, QQQ, IWM, IEF and GLD. Daily adjusted closes come from Yahoo Finance and proxy total returns. The sample starts at GLD's inception on 18 November 2004, the binding constraint, and ends 6 August 2026, giving 5,462 trading days.

Four macroeconomic series come from the St. Louis Fed's FRED database: the CBOE Volatility Index (VIXCLS), 2- and 10-year Treasury constant maturity yields (DGS2, DGS10) and the effective federal funds rate (DFF). I use DFF rather than FEDFUNDS because the former is daily and the latter a monthly average. The yield-curve slope is constructed explicitly as DGS10 minus DGS2 rather than taken from the published T10Y2Y series, which I retain only as a cross-check.

Alignment and missing values. The asset panel keeps only dates on which all five funds print an adjusted close. Macro series are reindexed to those trading days and forward-filled by at most five trading days; a longer gap raises an error rather than propagating a stale value. Backward filling is never used, so values before a series begins remain missing and are reported as such. Two calendar gaps longer than four days survive in the sample, both genuine exchange closures: the day of mourning for President Ford in January 2007 and Hurricane Sandy in October 2012.

Information availability. All FRED-sourced series enter signals with an additional one-trading-day lag, because H.15 yields are published with a lag and same-close availability of the VIX print is not guaranteed for a close-of-day decision. Price-derived features use information through the close of date t without extra lag. This policy is fixed for every specification.

Data licensing. Neither vendor permits redistribution of the raw series: Yahoo grants a personal, non-transferable licence, and VIXCLS carries a CBOE copyright requiring the owner's permission for non-personal use. The repository therefore publishes SHA-256 hashes and full metadata for every raw file, the download scripts, every generated table and figure, and per-phase provenance manifests, but no vendor data. A reader can audit every computation without receiving a byte of licensed data.

4 Preregistered empirical design

The hypothesis, comparator, sample split, cost convention and complete robustness grid were published and version-tagged before any model was estimated. Four amendments followed, each dated and each recording what was known when it was made; three were adopted before any portfolio return existed.

Hypothesis. The primary null is

$$H_0: \quad SR^{\text{net}}_{\text{regime}} - SR^{\text{net}}_{\text{rolling LW}} \leq 0, \tag{1}$$

tested net of 10 basis points in transaction costs. The comparator is deliberately the *closest non-regime strategy*, not equal weight or 60/40. Those naive benchmarks differ from the regime strategy in optimization, covariance estimation and regime conditioning simultaneously, so a gap between them would attribute nothing. Rolling Ledoit–Wolf minimum variance shares every ingredient except the regime signal, which isolates what regime conditioning adds.

Sample split. The initial training window runs from 18 November 2004 to 31 December 2009. The out-of-sample period begins 1 January 2010 and ends 6 August 2026. The first signal is formed at the December 2009 month-end close and executed on 4 January 2010, giving 200 monthly rebalancing decisions.

Information and execution timeline. At each month-end t: information is observed through the close of t; all models are estimated using data through t only; target weights are computed; trades execute at the close of the *next* trading day $t + 1$; and the new weights first earn the $t{+}1 \to t{+}2$ return. Because the dataset contains adjusted closes only, assuming execution at the same close where the signal was observed would constitute look-ahead. The convention costs one day of exposure and removes the ambiguity.

Look-ahead controls. A truncation test verifies that no derived quantity changes when future data is removed: for any date t, perturbing or deleting observations after t must leave every value dated t or earlier bit-identical. The test is applied to feature construction, volatility forecasts, covariance estimates and the full accounting layer.

Strategy ladder. Six strategies form an ablation ladder in which each rung adds exactly one ingredient: equal weight; 60/40 SPY/IEF; static minimum variance estimated once on the training window; rolling Ledoit–Wolf minimum variance; volatility-scaled minimum variance using EWMA variance forecasts with Ledoit–Wolf correlations; and regime-aware minimum variance. Reading performance down the ladder attributes any improvement to dynamic covariance, volatility forecasting or regime conditioning specifically.

Optimized strategies are long only, fully invested, and capped at 40% per asset, solved by sequential quadratic programming. The 60/40 benchmark is exempt from the cap by construction, since it deliberately holds 60% SPY. No strategy uses expected-return forecasts.

Costs. Turnover compares new targets against pre-trade holdings *drifted* by realized returns since the last rebalance, with the drift recurrence applied every trading day. Costs are charged on the full traded notional $\sum_i |w_i^{\text{target}} - w_i^{\text{pre}}|$; the halved figure is used only for reporting. Returns compound multiplicatively,

$$1 + r_t^{\text{net}} = (1 + r_t^{\text{gross}}) \left(1 - c \sum_i \left| w_{i,t}^{\text{target}} - w_{i,t}^{\text{pre}} \right| \right), \tag{2}$$

rather than by subtracting cost from return. Every strategy begins from 100% cash at the first execution, so all six pay an identical entry cost on a traded notional of one.

5 Volatility and regime estimation

Volatility. Three forecasts are compared out of sample: 63-day rolling historical variance, EWMA with $\lambda = 0.94$, and GARCH(1,1) with normal innovations refit at every rebalance date. The forecast target is the integrated variance over the next holding period, evaluated by QLIKE as the primary loss. GARCH attains the lowest QLIKE for all five assets, but only 1 of the fifteen pairwise comparisons survives Holm adjustment at the 5% level. EWMA was fixed *ex ante* as the model feeding the portfolio, so this comparison does not and cannot promote GARCH into the main specification; a GARCH-fed variant belongs to robustness.

Regimes. A two-state Gaussian hidden Markov model is refit at every rebalance date on an expanding window, using four standardized features: SPY log return, 21-day realized volatility, one-day-lagged log VIX and one-day-lagged yield-curve slope. Standardization uses the estimation window only.

The trading signal is the *filtered* probability $p_t = P(S_t = k \mid \mathcal{F}_t)$, taken as the posterior at the final observation of the sample truncated at t, where forward filtering and forward–backward smoothing coincide because no observation follows the endpoint. Full-sample smoothed paths are computed separately, stored in their own file and used only for descriptive figures; they never inform a trading decision.

Each refit runs sixteen predetermined initializations (seeds 42–57). States are relabeled after every fit by ascending training-window mean realized volatility, so state 0 is always the lower-volatility state; no code path assumes the estimator's raw labels carry meaning. A refit whose state occupancy falls below 5%, whose state covariance is singular, or whose transition matrix contains an absorbing row is flagged degenerate.

6 Covariance and portfolio construction

At each rebalance date the regime-conditioned covariance is built in four steps. First, the filtered state probability p_t. Second, horizon-averaged probabilities over the $H = 21$ day holding period, $\bar{p}_t(H) = H^{-1} \sum_{h=1}^{H} p_t P_t^h$, using the transition matrix estimated through t. Third, state-conditional covariances $C_{k,t}$, computed as responsibility-weighted covariances around each state's own mean and shrunk toward the unconditional Ledoit–Wolf estimate with weight $\alpha_k = n_{\text{eff},k}/(n_{\text{eff},k} + 60)$ on the state estimate, where $n_{\text{eff},k} = (\sum_s \gamma_{s,k})^2 / \sum_s \gamma_{s,k}^2$. Fourth, the mixture

$$\Sigma_{\text{RA},t} = \sum_k \bar{p}_{t,k}(H)\, C_{k,t}. \tag{3}$$

Equation (3) is a within-state mixture: it omits the between-state mean-dispersion term and so treats conditional means as common across states. That is a deliberate simplification consistent with the design's prohibition on expected-return forecasting, and it is documented as an assumption rather than presented as the complete law of total covariance. Its magnitude is measured at every rebalance: the omitted term reaches at most 0.16% of the within-state mixture and never exceeds 1%. The complete formula is carried as a robustness case.

Conditioning diagnostics motivate the shrinkage. Across 200 origins the raw EWMA covariance is the worst-conditioned estimator, averaging a condition number of 116 and peaking at 505, which is why the volatility-scaled strategy combines EWMA variances with Ledoit–Wolf correlations rather than using the EWMA covariance directly. The high-volatility state is materially worse conditioned than the low-volatility state (means of 61.4 versus 36.8), so the stressed-state matrix is nearest to singular exactly when a variance minimizer leans on it hardest. No matrix in the panel required a positive-semidefinite correction.

Degeneracy policy. A degenerate fit keeps its probability, which is reported unchanged, but does not supply a regime-conditioned covariance for that rebalance; the allocation uses the unconditional Ledoit–Wolf estimate instead, and the fallback is audit-logged. 4 of 200 origins invoked this rule, all for absorbing transition rows. The routing depends solely on the diagnostic flags; the affected dates are audit information and appear in no routing condition.

7 Primary results and inference

7.1 Realized performance

Table 1 reports the six strategies over 2010–2026, net of 10 basis points. The regime-aware strategy earned a Sharpe ratio of 0.929 against the comparator's 0.908. Reading down the ladder, most of the risk-adjusted gain over the naive benchmarks comes from optimization itself rather than from anything dynamic: equal weight earns 0.857 and static minimum variance 0.910, after which dynamic covariance adds nothing measurable (0.908) and the two responsive rungs add modest further amounts.

The highest realized Sharpe in the ladder belongs to the volatility-scaled strategy (0.945), not to the regime-aware strategy. That was not predicted by the design and is reported as found; it does not alter the preregistered hypothesis, which concerns the regime rung.

Table 1: Realized performance, 2010–2026, net of 10 basis points. Annualized figures; maximum drawdown from the net wealth path.

	CAGR (%)	Volatility (%)	Sharpe	Sortino	Max DD (%)	Calmar
Equal weight	11.61	11.90	0.857	1.214	−22.80	0.509
60/40	9.82	9.78	0.852	1.207	−21.20	0.463
Static min-var	9.04	8.23	0.910	1.301	−19.17	0.471
Rolling LW min-var	8.71	7.88	0.908	1.293	−17.41	0.500
EWMA-scaled min-var	8.70	7.54	0.945	1.356	−17.30	0.503
Regime-aware min-var	8.73	7.71	0.929	1.331	−17.44	0.501

7.2 The preregistered comparison

The estimated difference in annualized net Sharpe ratio between the regime-aware strategy and rolling Ledoit–Wolf minimum variance is

$$\widehat{\Delta SR} = 0.021, \qquad 95\% \text{ CI } [-0.075,\ 0.115], \qquad p = 0.327. \tag{4}$$

The interval is roughly 9.0 times as wide as the point estimate and contains zero comfortably; the estimate sits 0.437 bootstrap standard deviations from zero. H_0 is not rejected.

Inference uses a stationary bootstrap with 10,000 replications, mean block length 21 trading days and seed 12345, applied to *paired daily excess returns* annualized by $\sqrt{252}$ — the same estimand the point estimate reports. Both strategies receive identical bootstrap indices in every replication, preserving contemporaneous pairing; a test confirms that feeding the same series twice yields exactly zero difference with zero variance. The one-sided p-value comes from the centered null $\Delta SR_b - \widehat{\Delta SR}$, not from the fraction of ordinary draws below zero, which is a different quantity.

The conclusion is insensitive to block length: mean blocks of 10, 42 and 63 days all produce interval bounds within 0.005 of the primary bounds, and every one contains zero.

7.3 Returns and volatility separately

Mean returns are indistinguishable. Paired daily net return differences give an annualized arithmetic difference of 0.314 bps against a Newey–West standard error of 36.216 bps, so the estimate sits 0.009 standard errors from zero ($p = 0.993$, HAC lag 21; lags of 5 and 42 give the same conclusion).

Among the 5 secondary metric intervals, only annualized volatility excludes zero: the regime-aware strategy's is lower by 0.175 pp, with a 95% interval on that difference running from −0.327 pp to −0.033 pp. This is a secondary, unadjusted interval among several examined metrics and is not confirmatory evidence. What raises it above an incidental finding is that it is the outcome the mechanism predicts: a variance minimizer given additional information about variance should reduce variance. It is not described as statistically established.

7.4 Trading costs

The regime-aware strategy trades 54.19% annual half-turnover against the comparator's 16.94%, a factor of 3.2. At the primary cost assumption it spends 10.838 bps per year against the comparator's 3.388 bps, a hurdle of 7.450 bps that its risk reduction must clear. The estimated Sharpe advantage falls monotonically with costs, from 0.031 at zero to 0.011 at 20 basis points, and no cost level produces an interval excluding zero.

Table 2: Preregistered primary comparison. Paired stationary bootstrap, 10,000 replications, mean block 21 trading days, seed 12345, on daily excess returns.

	Δ Sharpe	CI lower	CI upper	p-value
Regime-aware − rolling LW	0.021	−0.075	0.115	0.327

8 Robustness

Thirteen specifications each vary one factor at a time from the primary model; no Cartesian product is constructed. Figure 1 shows the Sharpe difference and its interval across all of them, ordered as declared before any result was seen rather than sorted by effect.

Point estimates are positive in 11 of the thirteen specifications, and 12 of the thirteen intervals contain zero. **After Holm adjustment across the twelve-specification robustness family, nothing survives.**

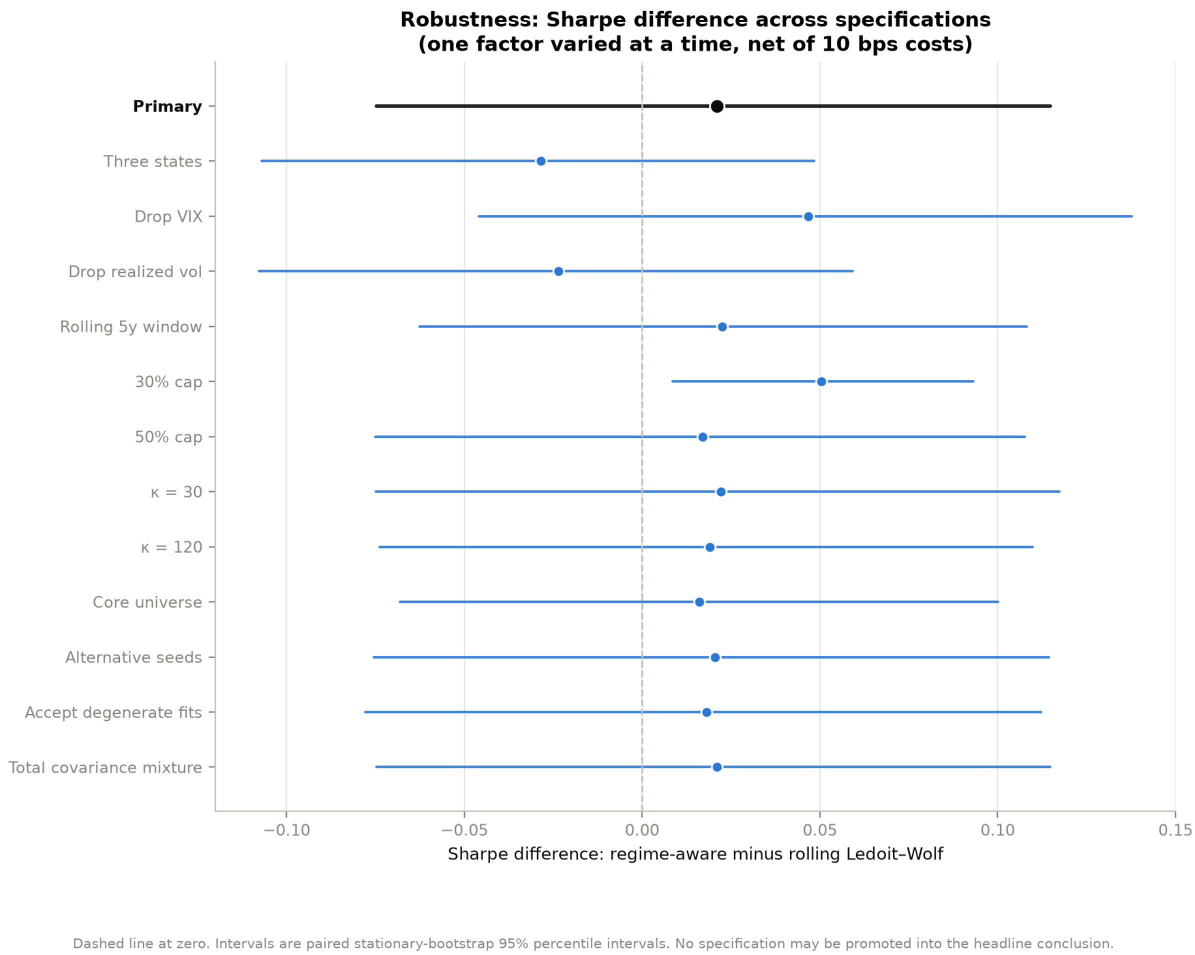

Figure 1: Sharpe difference between the regime-aware strategy and rolling Ledoit–Wolf minimum variance across thirteen specifications, each varying one factor. Intervals are paired stationary-bootstrap 95% percentile intervals; the primary specification is shown in black. Only the 30% weight cap excludes zero, and it does not survive Holm adjustment.

Sign instability. The estimate reverses sign in 2 of the preregistered variations: the three-state HMM (-0.028) and dropping realized volatility from the feature set (-0.023). Neither reversal is significant, but a result that changes direction when the state count changes is fragile. The three-state model is not itself defective — its middle state holds 45.4% occupancy with a minimum effective sample size of 487, canonical low–medium–high ordering holds at every refit, and no initialization failed — so the reversal reflects the estimate rather than a broken fit.

The weight cap. The 30% cap is the only specification whose unadjusted interval excludes zero (0.050, $p = 0.011$), and it does not survive Holm adjustment ($p = 0.133$). Decomposing it exactly, using $\Delta SR(0 \text{ bps}) - \Delta SR(10 \text{ bps})$ rather than approximating cost expenditure divided by volatility: reduced cost drag accounts for 0.008 of the 0.029 change from the 40% cap (26.6%), while gross pre-cost differences account for 0.022 (73.4%). Under the 30% cap the volatility advantage disappears while the relative turnover penalty falls substantially; however, reduced transaction costs explain only part of the larger Sharpe difference, with altered portfolio exposures and gross-return realizations accounting for the remainder.

Factors that barely move the estimate. The complete total-covariance mixture reproduces the primary estimate to four decimals, confirming the measured negligibility of the omitted between-state term. Shrinkage thresholds of 30 and 120 move the estimate by at most 0.002. An alternative seed block gives 0.020 against the primary 0.021, despite the selected seed changing at 152 of 199 consecutive refits. Accepting the degenerate fits rather than falling back gives 0.018.

Subperiods. Sliced descriptively from the primary paths, the pre-2020 estimate is 0.105 and the post-2020 estimate is -0.076. With 232 days in the COVID slice and 460 in the 2022 tightening slice, these samples are far too small for inference, and the sign disagreement between halves of the sample is a further reason for caution about the full-sample point estimate. No subperiod is promoted.

9 Limitations

Precision. Despite approximately 16.6 years of out-of-sample returns, the study had limited precision for detecting small differences in risk-adjusted performance. The estimated Sharpe difference was 0.021, while its 95% bootstrap interval ranged from -0.075 to 0.115. Consequently, the analysis provides no confirmatory evidence of improvement, but it also cannot exclude economically modest positive or negative effects. The results should therefore be interpreted as inconclusive regarding small effects rather than as evidence that regime conditioning has exactly zero value.

Table 3: Robustness grid: thirteen specifications, each varying one factor from the primary model.

	Δ Sharpe	CI lower	CI upper	Δ Vol (pp)
Primary	0.021	−0.075	0.115	−0.175
Three states	−0.028	−0.107	0.048	−0.180
Drop VIX	0.047	−0.046	0.138	−0.117
Drop realized vol	−0.023	−0.108	0.059	−0.052
Rolling 5y window	0.022	−0.063	0.108	−0.206
30% cap	0.050	0.009	0.093	0.033
50% cap	0.017	−0.075	0.108	−0.137
$\kappa = 30$	0.022	−0.075	0.117	−0.175
$\kappa = 120$	0.019	−0.074	0.110	−0.177
Core universe	0.016	−0.068	0.100	−0.188
Alternative seeds	0.020	−0.075	0.114	−0.175
Accept degenerate fits	0.018	−0.078	0.112	−0.171
Total covariance mixture	0.021	−0.075	0.115	−0.175

The binding constraint is information rather than raw observation count. Inference uses daily paired returns, but the strategy makes only about 200 rebalancing decisions, and serial dependence in both returns and regime states further reduces the effective information available to distinguish two highly correlated strategies.

Universe selection. Exchange-traded funds avoid single-name survivorship bias, but choosing these five funds in 2026 uses knowledge that they survived and remained liquid. All conclusions are conditional on the universe.

Vendor data. Yahoo adjusted closes are restated when distributions occur, so the analysis is tied to a specific frozen snapshot rather than to a canonical dataset. Cross-validation against CRSP or a commercial vendor would strengthen it.

Cost model. Costs are proportional to traded notional. The model excludes market impact, taxes, borrowing constraints and bid–ask dynamics, all of which would penalize the higher-turnover strategies further.

Scope. One market, one asset-class mix, one regime specification family. Whether the finding extends to international assets, credit, or intraday-informed volatility estimates is untested.

Numerical reproduction. Two independent runs produced filtered probabilities differing by up to 1.45×10^{-11}, while all selected seeds, classifications and reported conclusions reproduced. Exact argmax selection and tolerance-based selection were each stable across the two runs, although they selected different seeds at 144 of 200 origins. The evidence therefore supports numerical stability of both rules but does not establish that the tolerance rule was necessary for reproduction.

Researcher discretion. Preregistration, frozen data, causal tests, automated validation and dated amendments substantially constrained ex-post researcher discretion. All amendments were documented before viewing the results they could influence. Discretion was constrained, not eliminated: implementation choices were made throughout, and four amendments modified the frozen plan after it was tagged.

10 Discussion and conclusion

Regime conditioning modestly reduced realized volatility in the primary specification but did not produce statistically credible improvement in net risk-adjusted performance. Results were sensitive to state specification, feature selection, sample period, and portfolio constraints.

The mechanism is visible even though the effect is not. Conditioning a covariance forecast on an inferred volatility state does what theory suggests: the regime-aware portfolio realized 0.175 pp lower annualized volatility than an otherwise identical strategy. The difficulty is that this reduction is small relative to sampling variation, and harvesting it required 3.2 times the turnover, which consumed 7.450 bps per year more than its comparator. What survives after costs is 0.021 in Sharpe ratio, with an interval that runs from -0.075 to 0.115.

That pattern echoes DeMiguel et al. [2009], who found that none of fourteen optimization models consistently beat naive 1/N out of sample because estimation error offsets the theoretical gain from optimization. The result here sits one rung further up the same ladder: optimization beats naive allocation in this sample, but adding regime conditioning on top of dynamic covariance estimation does not produce a detectable further gain. The obstacle in both cases is the gap between what a model improves in principle and what survives estimation error and trading frictions in practice.

Two features of the robustness evidence deserve weight in any reading of the point estimate. The sign reverses under a three-state specification and when realized volatility is dropped from the feature set, and it reverses again between the pre- and post-2020 halves of the sample. A result that changes direction under preregistered variations of this kind is fragile regardless of its interval.

One observation the design did not set out to test deserves attention, and equally deserves careful labelling. Tightening the weight cap from 40% to 30% changed the estimated difference by more than any modelling choice examined, and decomposition attributes 73.4% of that change to altered exposures and realized returns rather than to the cost saving from reduced turnover. This is consistent with Jagannathan and Ma [2003], who show that constraints wrong as statements of belief can still improve realized risk by suppressing estimation error.

The status of that observation must be stated precisely, because it is the result most likely to be over-read. The cap sensitivity is consistent with constraints regularizing estimation error, but the specification was secondary, its interval was unadjusted, and it did not survive the robustness-family Holm correction. This result is hypothesis-generating rather than confirmatory evidence. What it suggests is methodological rather than empirical: studies of conditioning methods should treat constraint design as a first-order experimental factor rather than a fixed detail, since varying it here moved the estimate further than varying the regime model did.

Whether regime conditioning adds value in settings with more assets, higher-frequency rebalancing, or lower trading costs remains open. This study establishes only that in a five-asset ETF universe rebalanced monthly at realistic costs, over 16.6 years, the improvement was too small to distinguish from zero.

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